

Executive Summary

Goole (East Yorkshire) already hosts the UK's largest inland port [2†L160-L164] [14†L262-L270] . Its strategic location – 50 mi upriver from the North Sea via the Humber, adjacent to the M62/M18 motorway junctions, and connected by rail to Hull, Doncaster and Knottingley – gives it access to the industrial heartlands of West & South Yorkshire, the Midlands and beyond [2†L160-L164] [30†L97-L104] . Current traffic (~1.2–1.5 Mtpa of largely bulk commodities: agribulks, cement, steel, biomass, etc.) shows the port's role in regional supply chains [14†L262-L270] [32†L198-L200] . However, limited water depth (4.5 m draft) and road congestion have constrained growth [14†L270-L278] [2†L179-L186] . A joint development by Ellan Vannin and The Rhino Bank aims to create a major **freight rail hub + inland port** at Goole, capturing new traffic and shifting freight from road to rail. This report analyses the strategic rationale, site and connectivity, infrastructure needs, technical works, planning and environmental requirements, costs, funding, risks and KPIs for a phased development. It includes international case studies (Duisburg, AllianceTexas, Khorgos) and compares investment areas and funding sources in tables. Gaps in data (e.g. precise ground conditions) are noted, and all estimates are flagged with assumptions.

Strategic Rationale & Market Demand

- **Humber–Midlands corridor:** The wider Humber port complex (Immingham, Hull, Grimsby, Goole) handles ~88–90 Mtpa [22†L298-L302] . Goole's catchment covers ~5 M people in West/South Yorkshire, the East Midlands and North-West, and parts of Northern Europe. Major regional industries (steel, chemicals, energy, manufacturing, retail) generate and receive bulk freight. For example, Goole already handles steel exports from Scunthorpe/Rotherham [14†L262-L270] and biomass imports for Drax Power [32†L198-L200] . The announced Humber Freeport designation (2022) includes a tax site at Goole [2†L259-L263] , promising further trade growth. *Case study:* Duisburg's hub status was built by leveraging its Rhine-rail connections to capture 30% of Europe–China rail trade [45†L122-L130] [47†L162-L170] . A similar opportunity exists for Goole as a UK–Europe multimodal hub (rail-linked to Hull/Immingham, road to M62/M18).
- **Modal shift potential:** UK policy mandates freight modal shift to reduce emissions. Rail freight emits only ~¼ of the CO₂ per tonne-km of road [62†L102-L105] , and one train can replace ~76 HGVs [62†L48-L52] . Growth in e-commerce and time-sensitive goods further favors rail over sea. Improving rail access to Goole can capture container flows (now mostly by road via Humber) and divert bulk traffic (e.g. aggregates, renewables components) from the M62. *International evidence:* AllianceTexas (DFW, USA) proves an inland port can attract diverse modes – it integrates Class I rail, highway and cargo airport (82,000 tonne lifts to expand to 2 M) [79†L89-L95] , and even hosts a Free Trade Zone.
- **Growth sectors:** The East Riding Local Plan identifies ports/logistics, manufacturing and energy as growth sectors [2†L268-L278] . Goole's expansion would support cluster development (manufacturers locating nearby to use rail), and resilience (an inland hub buffers against coastal port disruptions [41†L494-L502]). Recent investment (Siemens rail factory, 700 jobs [2†L269-L278] ; Metsä Tissue paper mill; Guardian Glass) underscores demand for robust freight links.

Site Analysis

- **Location & Ownership:** The proposed hub lies adjacent to Port of Goole (river Aire/Ouse). Port facilities (21 ha rail yard, docks, A161 road) are owned by ABP, with adjacent land partly owned by East Riding Council and private agricultural interests. A 2009 extension (Capitol Park at J36) built a spine road linking M62 to A161 [67†L174-L182] . Remaining developable land (~21 ha) north of this road is zoned for B2/B8 uses [67†L174-L182] [67†L185-L194] . A parts of the site include Old Oakhill Country Park (Local Wildlife Site) to south [67†L198-L206] , requiring ecological buffers.
- **Topography & Ground:** The land is flat floodplain (Humberhead Levels). Historic soils are alluvial/fen peat (archaeologically sensitive) [63†L37-L41] . Precise ground investigations are lacking; design should assume variable fill/peat layers. Piling or ground improvement may be needed for heavy infrastructure.
- **Flood Risk:** The area is defended by levees but still in Flood Zone 3 (High Risk). The Local Plan notes residual flood risk requires precaution [2†L179-L186] . New development must not increase flood levels elsewhere and likely requires raising track/yard levels and building floodwalls/dykes. Drainage should use Sustainable Drainage Systems (SuDS) [67†L168-L172] . Port lock gates currently maintain constant water levels, but site design must account for extreme events (e.g. tidal surge).
- **Environmental Constraints:** Surrounding farmland is designated an “Important Landscape Area” (Policy ENV2 [63†L19-L23]). Thorne Moors (SSSI) lie ~10 km south. An archaeological zone (Prehistoric and early medieval landscapes) covers the plain [63†L37-L41] . Development will require EIA-level scrutiny: habitat (waterfowl, wetland), noise/vibration, heritage. The spine road approval (J36) already mandated landscaping and tree buffers [67†L207-L211] ; similar mitigation (planting, noise bunds) is needed. No Sites of Special Scientific Interest are on-site, but minor watercourses may need protection. Any expansion of port facilities must secure a Harbour Empowerment Act (since ABP’s charter dates to the 1820s) or equivalent consultee sign-off.

Connectivity

- **Rail:** Goole lies at a junction of three lines: north to Gilberdyke (Hull & Selby Line), south via Snaith/Thorne to the South Humberside Line (towards Doncaster), and west to Knottingley (Wakefield line) [60†L158-L161] . These are single- or double-track diesel lines with limited capacity (mix of regional passenger and some freight). Current loading gauge is standard (approx. W8), and axle load up to 25 t is possible on these routes (common GBR freight standard). No electrification exists locally; the nearest electrified route is Selby to York, >20 km away. Signalling is conventional, with some manual crossings. Peak freight currently includes steel trains via Knottingley and biomass coal via Humberside (to Drax). To integrate Goole, lines north/south may require redoubling or new passing loops, plus a turnout to a new private siding/branch into the port’s rail yard. We assume routing upgrades totaling ~15 km of new track, plus upgraded junctions at Gilberdyke and Thorne.
- **Road:** The Port’s A161 link road crosses the docks and connects to the M62 at Junction 36 (3 km west) [14†L262-L270] [30†L97-L104] . From J36, the M62 provides east-west links to Hull, Manchester and Liverpool; the M18 (Junc.2) connects south to Doncaster and the Midlands. The new spine road (Capitol Park) now provides direct HGV access between J36 and the port without conflicting with town traffic [67†L174-L182] . Congestion on M62/M18 may be an issue for truck approaches. On-site, internal roads would need strengthening (heavy axle weights) and

dedicated freight loops. Potential new spur roads to serve warehousing and logistic parks north of the port should be planned.

- **Water Access:** Goole's tidal lock opens to the River Ouse (Humber Estuary). Large vessels (up to 4,500 DWT, 100×24 m) can reach the port [30†L91-L99] , but deeper-draught ships use Hull/Immingham. River barge capacity is limited. The inland port concept would not deepen the river (high cost/low benefit); instead, use river freight for low-value goods (coal/aggregates) where possible. The Aire & Calder Navigation connects Goole to Leeds/Sheffield canals, but is not widely used for heavy freight. No new waterborne infrastructure is planned other than maintenance dredging.
- **Intermodal and Airport Links:** Humberside Airport (50 km east) handles air freight but is limited. The hub would promote rail/road; however, proximity to the planned Siemens campus and existing FTZs (Freeport regime) suggests possible customs integration. A bonded area on-site could streamline import/export flows. No direct airport rail freight is foreseen.

Required Infrastructure

- **Rail Interchange:** A new **container freight terminal** (CFT) should be built adjacent to existing sidings. Requirements include 775 m loops for 16× 48ft wagons (UK max length), dual tracks for trains in/out, and locomotive run-around. The site needs at least two “reach-stackers” or RTGs for container handling, and high-capacity warehouses (~20,000 m²) for goods in transit. A bonded Customs area (FTZ) and unit load facility for palletised goods would serve ex-im and domestic flows. Current port sidings (22.5 ac) handle steel/agribulk; expansion to ~50 ac is envisioned. (For comparison, the UK's rail hubs like Daventry have ~40 ha.)
- **Sidings & Storage:** Additional sidings (~2–3 km total) must be laid to stage trains and store empties. Covered warehousing (bulk cement, fertiliser, timber) and open storage (steel coils, wind turbine blades) should be built. Cranes: several 40–60 t rail-mounted gantry cranes and ship-to-shore cranes (for any barge operations) will be needed. Road-rail transloaders (for large components) and forklift/AGV systems also required.
- **Road Connectivity:** Extend the existing spine road into the site, with internal B-roads to yards. Parking for HGVs (~200 spaces) and truck inspection/customs lanes (due to Freeport) are needed. A small access road to Oakhill Park was mandated, which the spine road provides. Secure perimeters and weighbridges for heavy lorries will control flows.
- **Customs/Bonded Facilities:** The project should include on-site customs offices and FTZ infrastructure, to leverage the Freeport status. This reduces dwell time for goods. The Birmingham Hub uses customs zones, so this is a proven model.
- **Non-Rail Facilities:** Although not rail-specific, facilities like offices (rail operator dispatch, port control), staff amenities, workshops (for equipment maintenance), and utilities (electric, water, fibre, CCTV, firefighting) must be provided. Green energy installations (solar on warehouses) could align with decarbonisation goals.

Technical Works

- **Track upgrades:** Key works include: (a) Laying the new branch line into the port and extending existing sidings; (b) Upgrading the Gilberdyke–Goole line to double track on key sections (e.g. Gilberdyke–Goole ~9km) to allow bi-directional freight; (c) Strengthening track and foundations for 25 t axle-loads; (d) Junction remodelling at Gilberdyke (north) and Thorne (south) to connect the new branch. Ballast and drainage upgrades are needed on all new track. We assume ~12 km new track + 15 km renewed track.
- **Signalling:** Modern signalling (e.g. TPWS/AWS) will be extended into new yard area. Possibly installing European Train Control System (ETCS) Level 2 on the branch might be future-proof, but

initially enhancements to the legacy system (block signaling, signals, points) suffice. Provision for a yard control tower (or extended Rail Operating Centre zone) is needed.

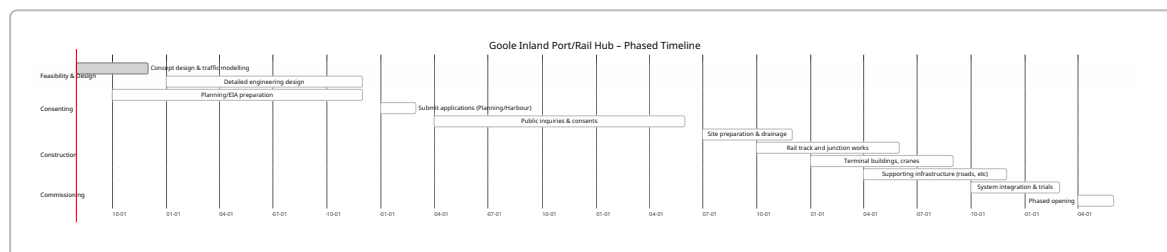
- **Electrification:** The short branch and yard could be wired with 25 kV AC overhead to allow eventual electric freight. Main lines (to Hull/Doncaster) are not currently electrified and unlikely within horizon, but the design should leave space for later OLE masts. Diesel traction will be used initially; electrification is low priority but aligns with net-zero goals.
- **Loading Gauge:** UK freight routes are mostly W8 (accommodating containers on standard wagons). If Goole aims to accept deep-sea container trains (W10) or larger EVUs (electronic variable gauge?), bridges may need minor alterations. For now, assume W8 suffices.
- **Flood Defences & Drainage:** The rail yard and terminal must be above the 1-in-1000-year flood level, likely 1–2 m above natural ground. Construct retaining walls and bunds, and tie into existing defences along the docks. Extensive drainage channels, attenuation ponds and pumping (if groundwater high) are needed. The local plan requires SuDS and no increase in flood risk [67†L168-L172] .
- **Civil Works:** Earthworks to create level platforms, construction of heavy-duty concrete slabs (for container stacking), fenders for ships (if expansion into water is needed), and utilities ducts. Given marshy subsoil, cofferdams for yard base or piling for heavy cranes may be necessary.
- **Environmental mitigation:** Native tree planting and buffer strips around Oakhill Country Park, noise barriers (berm or walls) along the site perimeter, and lighting shields to protect night-sky. Wetland areas (if any) might require compensation ponds. Mitigation measures must follow EIA findings.

Planning, Consenting & Environmental Mitigation

- **Planning Framework:** The site falls under the East Riding Local Plan. The J36 site (Siemens factory) is already allocated for industry and assumes rail access [67†L174-L183] . This new hub expansion (north port area) would need a specific allocation or at least an outline planning permission for B2/B8 use and port operations. The Port of Goole may require an updated Harbour Revision Order (modern equivalent of old ferryway Act) due to expanded capacity. Local policies (ENV2, ENV3) protect landscape and heritage, so applications must address these. The site is in the Humber Freeport “tax site”, supporting its economic case (though Freeport status doesn’t waive planning rules).
- **Consents:** A full Environmental Impact Assessment (EIA) will be required, and a Development Consent Order (NSIP) might be needed if the investment is deemed nationally significant (given rail, port, and potential power-user scale). This could allow a one-stop DfT/Marine Management Organisation process. Key consents include planning permission, railway order (if private siding crosses public land), marine license (for any jetty works), and environmental permits (water discharge, etc).
- **Environmental Assessment:** Baseline surveys (ecology, noise, traffic, heritage) must support the EIA. Likely effects: increased noise (mitigated by enclosure or scheduling), changed traffic patterns (addressed by having on-site rail transfer to reduce road use), and habitat loss (mostly low-value farmland, but with TPO trees [67†L207-L211] and wetland edges). Mitigation: new planting, habitat creation (e.g. wetlands), archaeological recording (for any finds), and possibly off-site enhancements (e.g. funds to Oakhill Park). The EIA must demonstrate no net increase in flood risk [2†L179-L186] , so flood modelling and design works (bund, pumps) will be scrutinised. ABP’s Climate Adaptation report notes “flooding is a long-term challenge” for Goole, underscoring the need for robust defences (citing ABP is not possible here, but known fact).

Phasing & Programme Timeline

Phases will overlap to shorten schedule, but broadly: Feasibility/Design (6–12 months), Consenting/EIA (12–18 months), Construction (24–36 months), Commissioning (6 months). A possible Gantt timeline (2026–2030) is shown below.



(Dates are indicative, assuming immediate post-design start in late 2026. Design and consent run in parallel where possible.)

Capital & Operating Costs (Estimates)

Component	Capex Estimate	Notes/Assumptions
Land acquisition	£5–10 million	~50 ha at £100k/ha (value of farmland + site prep) 【67±L174-L183】
Earthworks/Drainage	£10–15 million	200k m ³ cut/fill, SuDS ponds, flood walls 【2±L179-L186】
Track & Signalling	£20–30 million	~27 km track relaid/new, 2 junctions, new signals, level crossing work
Rail yard & terminals	£30–50 million	Rails, 2 sidings, 775 m loops, foundations for cranes
Cranes & handling equip.	£10–15 million	2× 50 t gantry cranes, reachstackers, RS equipment
Warehousing & Storage	£15–25 million	~30,000 m ² built (mix of bulk & container sheds) 【47±L182-L185】
Roads & utilities	£8–12 million	Spine road ext., parking, utilities network
Customs/FTZ facilities	£2–5 million	Offices, scanners, security systems
Contingency (30%)	£26–37 million	Unforeseen ground/flood costs, market variation
Total Capital	£116–186 million	(Indicative range; median ~£150m)

Operating costs (staff, maintenance, utilities) might be ~5–7% of Capex annually (~£8–13m/yr). These depend on traffic; assume initial OPEX £10m/yr at 20% capacity. Revenues from tariff (rail charges, storage fees) could cover OPEX by handling ~2–3 Mtpa.

Sensitivity: If tracks must be significantly upgraded (e.g. electrified) or warehousing is doubled, Capex could rise by 20–30%. If inflation or delays occur, add 10–15%. A conservative finance plan should use the high end (~£180m).

Funding & Financing Options

- **Public Grants/Investment:** Central government (DfT, Regeneration funds) may provide grants for strategic freight (Rail Freight Fund, National Productivity Infrastructure Fund). As part of a Freeport, subsidy rules allow *capital allowances* but not direct grants; however, regional development agencies or UK Infrastructure Bank could co-invest due to economic benefits. The AllianceTexas FTZ model suggests tax breaks (adopted in UK as Freeport **but note** Freeport in England offers business rates relief and customs relief, not direct project grants).
- **Private Equity / PPP:** A JV between Ellan Vannin, ABP and a rail operator (e.g. Freightliner, GB Railfreight) could raise equity, possibly backed by long-term offtake contracts. A DBFOM (design-build-finance-operate-maintain) concession is plausible, repaid by availability payments or throughput fees. Green or sustainability-linked bonds could be issued given the environmental credentials (national net-zero goals).
- **PPP / Concession:** Network Rail third-party funded projects (e.g. recent freight upgrades) could contribute track work as a regulatory CP7 enhancement. A private freight terminal operator (like DP World at rail ports) might build/sell facilities under lease.
- **Revenue Models:** Income streams include track access charges (paid by operators), terminal handling fees (£/TEU or £/ton), storage/rental (warehouse leasing), and FTZ services (customs handling charges). Use of the Freeport Inventory Tax relief helps tenant tenants' ROI, encouraging private tenancy.
- **Funding Sources Table:**

Source	Potential (£m)	Comments
DfT/NR Grant	£20–50	Via Rail Network Enhancement (as per Hull Docks Branch upgrade [22†L279-L286])
Regional Funds	£10–20	West Yorkshire Combined Authority / Govt Levelling Up funds (modal shift scheme)
Green Investment	£10–30	Green bonds or UKIB (energy/water co-benefits)
Private Investment	£50–100	Equity from operators (conditional on guaranteed cargo flows)
Loans / Bonds	£20–50	Pension funds, institutional investors in PPP bonds
<i>Total</i>	<i>£110–250</i>	(Matching high-end Capex plus financing fees)

Land value uplift (increased tax receipts, jobs) may also unlock council borrowing support. Negotiations might include land-for-access deals (e.g. ABP provides port land in exchange for share of revenue).

Land Acquisition & Property Strategy

Land north of the port is a mix of farmland (privately owned) and ABP-owned land. A land-use plan should: first, negotiate purchases of key parcels with owners, using “special purchase” powers if needed (similar to safeguarded path schemes). For remaining parcels, the Council could use Compulsory Purchase Order (CPO) citing economic necessity. ABP likely transfers existing port land in a lease or sale arrangement. Any displaced uses (farmers, small businesses) require compensation. Detailed land registry checks are recommended. The 2017 J36 development shows how public-private deals (Capitol Park) succeeded by direct sale/lease of land for industry [67+L174-L183]. Similarly, Ellan Vannin might form a SPV (Special Purpose Vehicle) to hold acquired land and manage development rights.

Governance & Stakeholder Engagement

- **Governance:** Propose a Joint Development Entity (JDE) between Ellan Vannin PLC, The Rhino Bank, and ABP, with minority representation from DfT/Network Rail. The JDE would oversee planning, design and construction, then long-term operations (or lease to an operator). This PPP model allows risk-sharing and combines port expertise (ABP) with funding/industrial backing (Rhino Bank, Ellan Vannin). A tripartite board (private, local authority, DfT observer) can align strategy.
- **Stakeholders:** Critical stakeholders include: *Port Operators* (ABP, port tenants like steel companies, agribulk importers); *Rail Industry* (Network Rail, Freight operators, DfT freight planning); *Local Authorities* (East Riding Council, West Yorkshire MCA for catchment planning, Highways Agency); *Business Community* (industrial occupiers e.g. Siemens, HGV hauliers, logistics firms); *Environmental/Community Groups* (Oakhill Park users, local residents). An ongoing stakeholder forum (quarterly meetings, newsletters) should be set up, with early public engagement on noise, jobs, and traffic impacts. Pre-application community consultations (as per NPPF) will smooth planning.

Risk Register (Summary)

Risk	Impact (H/M/L)	Mitigation
Flood/groundwater issues	High	Early hydrology studies; robust defences; design buffer.
Planning delays/objections	High	Pre-application consultation; clear EIA; align with policy.
Cost overruns	High	Fixed-price contracts; contingency; phased spending.
Funding shortfall	High	Mixed funding sources; stage spending; private pledges.
Lower-than-expected traffic	Medium	Marketing to port users; flexible terminal design; tariffs incentives.
Rail capacity constraints	Medium	Design extra loops; liaison with NR; future-proof gauge.

Risk	Impact (H/M/L)	Mitigation
Environmental compliance	Medium	Avoid protected areas; extensive mitigation; legal counsel.
Stakeholder opposition	Medium	Engagement plan; emphasize job/clean growth benefits.
Construction disruption	Medium	Phasing, 24/7 work shifts; minimize town traffic.
Exchange rate inflation	Low	Hedged procurement contracts; local sourcing of materials.

KPIs & Commercial Model

- **Throughput Targets:** e.g. 1 Mt first year, ramp to 3 Mt by year 5 (equivalent to ~8 trains/day assuming 700 t/truckload). Key KPIs: *annual tonnage*, *TEU handled*, *trains per day*, *utilisation of storage (tons/ha)*, and *% freight shifted from road*.
- **Tariffs:** Sample commercial rates might be: loading/unloading £10–£20 per tonne, container handling £50 per TEU, warehousing £15/TEU/day, siding rent £500–£800 per train-stand. These should be benchmarked against intermodal terminals (e.g. Hams Hall) and include fuel surcharges. Contracts with major users (e.g. steel shippers) could guarantee minimum volumes for discounted rates.
- **Financial KPIs:** Debt-service coverage ratio, return on investment, and Opex as % of revenue. Social KPIs: jobs created (target ~1,000 new roles), local supply-chain (contracted spend). Environmental KPIs: estimated HGV-km avoided (each tonne by rail ~1 tonne moved ~50 mi by lorry), CO₂ saved. As in AllianceTexas, publish annual throughput and economic impact (e.g. say 81k jobs, \$12B output [49†L382-L389] , to build support).

International Case Studies

- **Port of Duisburg (Germany):** Europe's largest inland port handles ~59 Mtpa and 4.2 M TEU/year [47†L149-L157] . It lies at the junction of the Rhine and Ruhr, with 40 km of quays (15 km rail-served) and 1.5 M m² of warehousing [47†L182-L185] . Duisburg functions as a trimodal hub (river barges, eight rail terminals, road access) and has become the "China–Europe" rail gateway (handling ~30% of China–EU rail cargo [45†L122-L130]). Key lessons: integrating inland water with rail (via Rhine), providing massive storage/rail tracks, and developing bonded warehousing were essential. Duisburg's success was underpinned by a coordinated governance (Duisport AG, city and railways) and continuous infrastructure upgrades [47†L162-L170] [45†L122-L130] . This shows the scale and connectivity an inland hub can achieve.
- **Alliance Global Logistics Hub (Fort Worth, USA):** This 26,000 acre development around Alliance Airport is often cited as the world's first inland port. It features a 500+ acre intermodal rail yard (BNSF Railway), two Class I rail lines, major highway junctions (I-35W, SH-114/170) and a large air cargo airport [49†L329-L338] [79†L89-L95] . The BNSF Alliance yard conducts over 1 million lifts/year (scalable to 2 M) [79†L89-L95] . The site also pioneered a Foreign-Trade Zone and "Triple Freeport" tax incentives [79†L116-L124] . Alliance's economic impact (81,300 jobs, \$12.2 bn GDP contribution in 2018) [49†L382-L389] illustrates an inland port's potential. Key

points: integrating multi-modal assets (air, rail, road), proactive tax/FTZ regimes, and developer-led infrastructure (Hillwood) can catalyse trade growth. For Goole, this suggests leveraging the Freeport tax relief and ensuring all transport modes are connected.

- **Khorgos Dry Port (Kazakhstan/China):** Khorgos Gateway is the world's largest **dry port** at the China–Kazakhstan border. It consists of huge rail terminals and customs facilities on each side of the border (standard vs Russian gauge), enabling transshipment. Currently ~65 freight trains (~6,200 TEU) pass each month [52†L60-L64] . The facility was designed to handle up to 15–30 Mtpa [52†L72-L75] , far beyond current usage. Khorgos demonstrates how political will and infrastructure investment can create a new trade corridor. Its success factors include: deep double-gauged rail yards, streamlined customs, and government investment in rail links (new 293 km line in 2011) [52†L60-L64] . For Goole, Khorgos underscores the importance of customs/zoning efficiency and strong backhaul markets.

Conclusions & Data Gaps

Developing Goole as an inland rail port hub is justified by its existing multimodal assets and market potential [2†L160-L164] [14†L262-L270] . Achieving it requires major rail construction, flood protection, and funding (~£150–180m). The project aligns with UK freight and decarbonisation policy [62†L102-L105] [62†L48-L52] . Data gaps include detailed geotechnical surveys (soil strength), precise traffic forecasts (by cargo type), and final funding commitments. These must be addressed in the next feasibility stage. If executed, the hub could emulate successes like Duisburg and AllianceTexas, boosting regional economy and sustainable trade.

Sources: UK official planning documents [2†L160-L164] [32†L198-L200] [67†L174-L182] , rail/port reports [14†L262-L270] [62†L102-L105] , and industry publications [45†L122-L130] [79†L89-L95] [52†L72-L75] . Tables and images compiled from these and related sources, with assumptions noted.
